# Detecting and steering object hallucination from hidden states

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## Evidence status

This document accompanies a verified saved run. Read the generated results/REPORT.md and results/REPORT_metrics.json for the complete measured tables and confidence intervals.

## Research question

Can a terminal-prompt decoder state predict an incorrect answer on unseen images, and can norm-preserving activation rotation reduce false-yes answers while retaining recognition of present objects? The motivating false-yes question is narrower than the active detector endpoint: the configured detector predicts any wrong answer, including a wrong no.

## Frozen image partitions

| Partition | Full profile | Quick profile | Used for |
|---|---|---|---|
| calibration | 100 | 40 | Layer, probe, direction, gate |
| validation | 50 | 20 | Intervention strength |
| test | 350 | 100 | Reporting only |

Source: dataset/pope/splits_seed20261030.json and configs/pope_llava.json. These counts are planned image assignments, not evidence that inference completed.

## Readout and intervention

Capture raw decoder block outputs at the last prompt position before generation, excluding embeddings. Select the linear-probe, mean-difference and PCA-projection layers using image-grouped calibration folds. Answer-token negative log probability and entropy are later first-token baselines, not early-warning features.

Orient the direction toward correct calibration answers. Choose cosine gates on calibration and intervention strength on validation. SLERP preserves activation norm, whereas addition can change it. Random-direction and shuffled-label controls reuse the selected SLERP strength. Norm preservation alone establishes no factual benefit.

## quick profile: 100 held-out test images

Run a57da9f4810959cc | Tesla T4 | Target: error

The layer 13 error-detection probe reached AUROC 0.719 [0.667, 0.769]; its paired improvement over answer-token confidence is inconclusive. SLERP changed false-yes rate by -0.007 [-0.016, -0.001] and recall by -0.020 [-0.037, -0.007]; superiority to both matched controls is inconclusive.

## Unsteered baseline

| POPE split | Accuracy [CI] | Recall [CI] | F1 [CI] |
|---|---|---|---|
| random | 0.882 [0.860, 0.905] | 0.787 [0.743, 0.827] | 0.869 [0.842, 0.897] |
| popular | 0.865 [0.840, 0.890] | 0.787 [0.743, 0.827] | 0.854 [0.823, 0.882] |
| adversarial | 0.843 [0.815, 0.872] | 0.787 [0.743, 0.827] | 0.834 [0.803, 0.865] |
| pooled | 0.863 [0.840, 0.888] | 0.787 [0.743, 0.827] | 0.852 [0.824, 0.880] |

## Detection on unseen images

| Detector | AUROC [CI] | AUPRC [CI] |
|---|---|---|
| Linear probe | 0.719 [0.667, 0.769] | 0.268 [0.208, 0.349] |
| Mean difference | 0.783 [0.745, 0.823] | 0.320 [0.250, 0.411] |
| PCA projection | 0.584 [0.522, 0.641] | 0.175 [0.133, 0.227] |
| Answer-token -log p | 0.745 [0.683, 0.804] | 0.350 [0.266, 0.448] |
| Answer-token entropy | 0.745 [0.683, 0.804] | 0.347 [0.264, 0.447] |

## Steering and matched controls

| Method | False-yes delta [CI] | Recall [CI] |
|---|---|---|
| SLERP / PCA | -0.007 [-0.016, -0.001] | 0.767 [0.723, 0.810] |
| Addition | 0.002 [0.000, 0.006] | 0.793 [0.750, 0.833] |
| Random direction | -0.001 [-0.003, 0.000] | 0.787 [0.743, 0.827] |
| Shuffled labels | -0.004 [-0.013, 0.000] | 0.773 [0.727, 0.817] |

All intervals are image-clustered 95% bootstrap intervals. Rates use the 0-1 scale. Read the full generated report for precision, yes-ratio, retention, validation diagnostics and paired control comparisons.

Source: results/REPORT_metrics.json and results/published_run.json. No weights or source COCO pixel bytes were independently rehashed from this return bundle.

## Scoring and uncertainty

Use the official pinned POPE yes/no parser and keep all questions about an image in the same partition, across question sets. Report false-yes rate on absent objects, recall on present objects, and retention among baseline-correct answers as distinct quantities.

Use 1000 image-clustered bootstrap resamples with seed 20261030. Paired comparisons resample the same images across methods. Intervals are conditional on the frozen calibration and validation choices; they do not establish generalisation beyond the benchmark.

## Declared interpretation rules

Detection improvement requires a wholly positive paired interval for probe AUROC minus answer-token negative-log-probability AUROC. Geometry requires the held-out PCA-projection AUROC interval to lie above chance, without implying a universal truth axis. Steering reduction requires a wholly negative paired false-yes delta; recall loss must be stated alongside any reduction. Undefined or crossing intervals remain inconclusive.

## Corrections and limitations

Pixel review retained 16 of 50 custom benchmark rows. The earlier scorer allowed an unrelated negation to override an affirmative absent-object claim. Earlier headline detection and reduction claims are withdrawn because complete valid held-out evidence was not recovered. The corrected custom set is secondary; proposed relabels are not applied.

POPE is a binary object benchmark, not a general test of long-form hallucination, video, audio or deployed safety. A quick run is exploratory. No external user study, enterprise-risk reduction or production safety benefit is claimed.

## Modern-model replication

A separate baseline and detection-only replication is registered for Qwen/Qwen2.5-VL-7B-Instruct, revision cc594898137f460bfe9f0759e9844b3ce807cfb5. Status: Not executed. The same questions, scorer and splits are required. The adapter is stub-tested; actual checkpoint execution and T4 memory fit remain unverified.

## Reproduce

Open notebooks/CounterPrior_Mechanistic_Steering.ipynb on branch claude/site-2026-11 in Colab, select a GPU and run the setup, timing, experiment and report cells. Run files stay on Drive. Keep the configuration fixed after test evaluation; report every executed profile, including negative results.

CLI: bash scripts/colab_run.sh setup; then use a fixed COUNTERPRIOR_RUN and COUNTERPRIOR_PROFILE=quick for timing, run and report. The headless plotting backend is forced to Agg so Colab cannot pass an unavailable inline backend into the isolated environment.

Code: https://github.com/basavarajnaduvinamani/DA5410-Winter-Project/tree/claude/site-2026-11

Sources: docs/PROTOCOL.md; configs/pope_llava.json; dataset/audit_2026-10-30.json; site/src/data/research.json; results/REPORT.md.